FEETFLOW

INTELLIGENT FLEET OPERATIONS PLATFORM

UI / UX DESIGN DOCUMENT

Visual Language, Component Library & Screen Specifications

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1. Design Philosophy & Vision

1.1 Design Direction

FEETFLOW adopts a "Dark Command" visual aesthetic — inspired by high-stakes logistics dashboards, drone delivery platform UIs, and military-grade control centers. The interface communicates authority, precision, and real-time intelligence while remaining intuitive for daily operational use.

1.2 Design Pillars

Pillar	Description	Application
Command Clarity	Every screen answers one primary question at a glance	Dashboard KPI hierarchy, status pills
Data Density with Breathing Room	High-information layouts with intentional whitespace	Tables, cards with 24px gutters
Trust through Color	Status always communicated via color + icon — never color alone	Vehicle states, alert severity levels
Motion with Purpose	Micro-animations confirm actions, never decorate	State transitions, loading indicators
Modular Scalability	Components scale from 10 to 10,000 vehicles without redesign	Responsive grid, virtualized tables

1.3 Design Inspiration

The visual language draws from drone delivery platform concepts featuring: deep dark backgrounds (navy/black) that reduce eye strain during extended operational shifts, electric blue and neon green accents that instantly draw attention to critical data, card-based layouts that create clear visual hierarchy between modules, and bold sans-serif typography that remains legible at all screen densities.

2. Brand Identity

2.1 Logo Usage

Variant	Usage Context	Minimum Size	Background
Full Logo (Icon + Wordmark)	Login screen, onboarding, reports	160px wide	Dark navy or white
Icon Only	Collapsed sidebar, favicon, app icon	32px × 32px	Any
Wordmark Only	Document headers, export footers	120px wide	Light backgrounds
Monochrome White	Dark overlays, loading screens	120px wide	Dark only

2.2 Logo Clear Space

Maintain clear space equal to the height of the "F" letterform on all sides of the logo. No other visual elements — text, icons, or imagery — may enter this exclusion zone.

2.3 Brand Voice in UI Copy

Attribute	DO	DON'T
Tone	Direct, confident, operational	Casual, playful, vague
Error Messages	"Vehicle capacity exceeded. Reduce cargo by 200 kg."	"Something went wrong."
Success States	"Trip dispatched. Driver and vehicle status updated."	"Done! Great job!"
Empty States	"No active trips. Dispatch a vehicle to begin."	"Nothing here yet!"
Alerts	"Insurance expires in 7 days — action required."	"Heads up! Check insurance."

3. Color System

3.1 Core Palette

Token	Hex	Role	Usage
color-bg-base	#0A0F1E	Primary Background	App shell, sidebar, main canvas
color-bg-card	#111827	Card Background	Modules, panels, modals
color-bg-elevated	#1E2D40	Elevated Surface	Dropdowns, tooltips, drawers
color-accent-electric	#00C2FF	Primary Accent	CTAs, links, active states, borders
color-accent-lime	#39FF14	Success / Online	Available status, positive KPIs
color-accent-orange	#FF6B35	Warning / Alert	License expiry, cost overruns
color-accent-red	#FF3B55	Danger / Critical	Blocked vehicles, expired docs
color-text-primary	#E8EDF5	Primary Text	Headings, labels, key data
color-text-secondary	#94A3B8	Secondary Text	Descriptions, metadata, timestamps
color-border	#1E3A5F	Border	Card outlines, table dividers

3.2 Semantic Status Colors

Status	Color Token	Hex	Icon	Use Case
Available	color-accent-lime	#39FF14	checkmark-circle	Vehicle/driver ready
On Trip	color-accent-electric	#00C2FF	navigation	Currently dispatched
In Shop	color-accent-orange	#FF6B35	wrench	Under maintenance
Blocked	color-accent-red	#FF3B55	ban	Compliance violation
Idle	#F59E0B	#F59E0B	clock	Engine on, no movement
Offline	#64748B	#64748B	wifi-off	No telemetry signal

3.3 Light Mode (Reports & Exports)

When generating PDF and Excel exports, the system switches to a light-mode palette to ensure print legibility. Background becomes #FFFFFF, text shifts to #0A0F1E, and accent colors remain consistent. The Command Center Dashboard remains dark-mode only.

4. Typography Scale

4.1 Font Family

Role	Family	Weights Used	Fallback
Display / Hero	Inter	700, 800	Arial, Helvetica
Headings	Inter	600, 700	Arial, Helvetica
Body / UI Labels	Inter	400, 500	Arial, Helvetica
Data / Numbers	Inter (tabular nums)	500, 600	Arial Narrow
Code / Log Values	JetBrains Mono	400	Courier New

4.2 Type Scale

Level	Size	Weight	Line Height	Usage
Display	48px / 3rem	800	1.1	Hero KPIs (e.g. total fleet count)
H1	32px / 2rem	700	1.2	Page titles (e.g. "Vehicle Registry")
H2	24px / 1.5rem	700	1.3	Module section headers
H3	20px / 1.25rem	600	1.4	Card titles, sub-sections
H4	16px / 1rem	600	1.4	Table column headers
Body L	16px / 1rem	400	1.6	Primary body text
Body M	14px / 0.875rem	400	1.5	Table rows, secondary content
Caption	12px / 0.75rem	500	1.4	Timestamps, metadata, tooltips
Label	11px / 0.6875rem	600	1.3	Status pills, badges (uppercase)

4.3 Typography Rules

- Use tabular number variant (font-variant-numeric: tabular-nums) for all KPI values and data tables to maintain alignment.
- Status pill labels are always uppercase with letter-spacing: 0.08em.
- Never exceed 75 characters per line in body text blocks.
- KPI display numbers use the Display size with a secondary unit label at H3 (e.g., "1,284 vehicles").

5. Component Library

5.1 Status Pills

Status pills are the primary communication mechanism for vehicle and driver states throughout FEETFLOW. They are always pill-shaped (border-radius: 9999px) with a colored left border, icon, and label.

Pill Variant	Background	Text & Border Color	Icon	Min Width
Available	#0D2E14	#39FF14	check-circle	96px
On Trip	#0B2340	#00C2FF	navigation	80px
In Shop	#2D1A0A	#FF6B35	tool	80px
Blocked	#2D0A12	#FF3B55	slash	76px
Idle	#2D240A	#F59E0B	clock	64px
Expired	#2D0A12	#FF3B55	alert-triangle	80px

5.2 KPI Cards

KPI cards appear on the Command Center Dashboard in a 4-column responsive grid. Each card contains: a metric label, a large display number, a trend indicator (up/down delta), and a sparkline chart. Cards have a subtle left accent border in the relevant status color.

Card Element	Spec	Notes
Container	280px × 160px min, flex-grow	Background: color-bg-card, border-radius: 12px
Accent Border	4px solid [status color]	Left side only, full card height
Metric Label	Caption size, color-text-secondary	Uppercase, letter-spaced
Primary Value	Display size, color-text-primary	Tabular nums, always visible above fold
Delta Badge	Label size, positive=lime / negative=red	Arrow icon + percentage
Sparkline	100% width × 40px, Chart.js area	Last 30 days, accent color fill
Padding	20px all sides	Minimum 16px for small viewports

5.3 Data Tables

All fleet data tables share a standardized structure: a sticky header row, sortable columns, row hover states, checkbox multi-select, and inline action buttons. Tables support bulk operations for vehicle and driver management.

Table Element	Spec
Header Row	Background: #0D1B2A | Text: color-accent-electric | Height: 48px
Body Row (even)	Background: #111827 | Height: 52px minimum
Body Row (odd)	Background: #0F1B2D | Height: 52px minimum
Row Hover	Background: #1A2F4A | Transition: 120ms ease
Row Selected	Background: #0D2340 | Left border: 3px solid electric
Cell Padding	12px horizontal, 0 vertical (vertically centered)
Inline Actions	Icon buttons (24px): Edit, View, Delete — visible on row hover
Sort Indicator	Up/down chevron in header, electric color when active

5.4 Alert & Notification Components

Type	Color	Icon	Behavior
Info	color-accent-electric	info	Persists, manual dismiss
Success	color-accent-lime	check-circle	Auto-dismiss 4s
Warning	color-accent-orange	alert-triangle	Persists, manual dismiss
Critical	color-accent-red	alert-octagon	Persists + audio cue, requires action
Rule Trigger	color-accent-orange	zap	Links to rules engine, shows triggered condition

5.5 Form Elements

Element	Default State	Focus State	Error State
Text Input	Border: color-border, bg: color-bg-elevated	Border: electric, glow shadow 0 0 0 3px rgba(0,194,255,0.2)	Border: red, error text below
Select / Dropdown	Same as input + chevron icon	Same as input focus	Same as input error
Toggle Switch	Off: #374151, On: electric	Outline focus ring: electric 2px	N/A
Checkbox	Outline: color-border	Checked: electric fill + white check	Outline: red
Date Picker	Calendar icon suffix	Opens date popover	Highlights expired dates in red

6. Layout Grid System

6.1 App Shell Structure

Zone	Dimensions	Content	Behavior
Top Navigation Bar	Full width × 56px	Logo, global search, alerts bell, user avatar	Fixed, z-index: 1000
Left Sidebar	260px (expanded) / 64px (collapsed)	Module navigation, role-based links	Collapsible, persists preference
Main Content Area	Fluid fill (sidebar-aware)	Page-specific modules and tables	Scrollable, 24px padding
Right Context Panel	320px (optional)	Trip details, driver profile drawer	Slides in on selection, dismissible
Notification Tray	400px × full height	Real-time alerts, rule triggers	Overlay, right-anchored

6.2 Content Grid

The main content area uses a 12-column grid with 24px gutters and 24px padding. Dashboard cards span 3 columns (25%) on large screens, collapsing to 6 columns (50%) on medium and 12 columns (100%) on mobile.

Grid Property	Value
Columns	12
Gutter Width	24px
Content Padding	24px (desktop), 16px (tablet), 12px (mobile)
Max Content Width	1440px (centered on ultra-wide displays)
Base Spacing Unit	4px (all spacing is multiples of 4)

7. Screen-by-Screen Specifications

7.1 Login Screen

Element	Specification
Background	Full-viewport dark navy (#0A0F1E) with subtle animated particle grid
Layout	Centered card (480px wide), vertically centered, 48px padding
Logo Position	Top of card, centered, 120px wide
Headline	"Fleet Intelligence. Redefined." — H2 size, white
Email Input	Full width, 48px height, standard form spec
Password Input	Full width, 48px height, show/hide toggle
CTA Button	"Sign In" — full width, 48px height, electric blue fill
Forgot Password	Caption text link below CTA, electric color
Role Indicator	Small badge appears after authentication showing role (Fleet Admin, Driver, etc.)
Error Handling	Inline shake animation on invalid credentials, red error text below inputs

7.2 Command Center Dashboard

The primary landing screen after login. Provides a real-time operational overview for Fleet Managers and Operations Managers.

Zone	Content	Data Source
KPI Row (top)	4 cards: Total Vehicles, Active Trips, Drivers Available, Monthly Fuel Cost	Live DB aggregation
Fleet Health Map	Route map or zone map showing vehicle positions (Phase 2: live GPS)	Trip records + GPS API
Alert Feed	Right panel: recent rule triggers, license expiry warnings, cost alerts	Rules Engine
Maintenance Due	Table: next 10 vehicles due for service (sorted by urgency)	Maintenance records
Revenue vs Cost	Line/area chart, last 30 days, two overlapping series	Financial module
Driver Leaderboard	Top 5 drivers by performance score, sparkline per driver	Driver scoring engine

7.3 Vehicle Registry

Full CRUD interface for fleet vehicles. Accessible by Fleet Admin and Operations Manager roles.

Element	Specification
Page Header	"Vehicle Registry" H1 + vehicle count badge + "Add Vehicle" primary CTA
Filter Bar	Status filter pills (All / Available / On Trip / In Shop / Blocked) + search input
Vehicle Table Columns	Vehicle ID, Make/Model, Year, Plate, Status Pill, Capacity, Last Service, Actions
Add/Edit Drawer	640px right-side slide panel with vehicle form fields
Document Upload	RC, Insurance, Permit — file drop zone, shows expiry date after upload
Status Auto-Logic	Status pill auto-updates based on trip dispatch and maintenance assignment
Delete Confirmation	Modal dialog with vehicle name, "This action cannot be undone" warning
Bulk Actions	Multi-select checkboxes enable: Export CSV, Archive, Change Status

7.4 Trip Dispatcher

The core operational screen where trips are created, validated, and dispatched. Used primarily by Dispatchers.

Step / Element	Specification
Step 1: Route Details	Origin (autocomplete), Destination (autocomplete), Scheduled departure
Step 2: Cargo Info	Cargo weight (kg), cargo type, cargo value (optional)
Step 3: Vehicle Selection	Filtered list showing only AVAILABLE vehicles + capacity vs cargo weight check
Validation — Cargo	If cargo weight > vehicle capacity → red banner: "Cargo exceeds capacity by X kg. Select a larger vehicle."
Step 4: Driver Assignment	Filtered list showing only AVAILABLE drivers + license expiry check
Validation — License	If license expired → driver card disabled + tooltip: "License expired on [date]. Cannot assign."
Dispatch Button	Enabled only when all validations pass. Electric blue, "Dispatch Trip" label.
Post-Dispatch	Success toast + vehicle status → On Trip + driver status → On Trip

7.5 Maintenance & Service Logs

Element	Specification
Log Entry Form	Vehicle ID (dropdown), service type, cost (₹), service date, service provider, notes
Status Auto-Trigger	Adding maintenance entry → vehicle status auto-set to "In Shop"
Completion	"Mark Complete" button → vehicle returns to Available status
Service History Table	Per-vehicle timeline view: date, type, cost, provider, tech notes
Reminder System	Auto-alerts appear in notification tray when next service is within 7 days or mileage threshold

Element	Specification
Cost Rollup	Maintenance cost per vehicle visible in vehicle detail and financial reports

7.6 Fuel & Expense Logger

Element	Specification
Entry Fields	Vehicle ID, trip ID (optional link), fuel amount (L), cost (₹), date, odometer reading, pump/vendor
Auto-Calculation	Cost per km = total fuel cost ÷ km driven (auto-computed from odometer delta)
Expense Categories	Fuel, Toll, Parking, Driver Allowance, Repair, Miscellaneous — color-coded
Threshold Alerts	Rule Engine: if fuel cost > configured limit → orange alert in notification tray
Monthly Summary	Bar chart: expense by category, month-over-month comparison
Export	CSV or PDF with date range filter for accountant/finance manager role

7.7 Driver Safety & Performance Profiles

Element	Specification
Driver Card	Avatar/photo, name, ID, current status pill, license expiry countdown, overall score badge
Performance Score	Composite score (0–100): trip completion rate, speed compliance, incident rate
Score Breakdown	Gauge chart showing sub-scores for: Speed, Punctuality, Incidents, Fuel Efficiency
Trip History	Scrollable table: date, route, duration, distance, fuel used, notes
License Tracker	Document chip showing license type, number, expiry — red when <30 days
Incident Reports	Driver can submit via mobile app; manager reviews and closes in this panel
Penalty Flags	Rule: 3 over-speed alerts → flag badge appears on driver card, notification sent to Safety Officer

7.8 Analytics & Reports

Report Type	Chart Type	Filters	Export
Fleet Utilization	Stacked bar (active/idle/in-shop per vehicle)	Date range, vehicle group	CSV, PDF
Financial Summary	Area chart (revenue vs cost) + P&L table	Month, vehicle, category	CSV, PDF
Driver Performance	Leaderboard table + radar chart per driver	Date range, team	CSV, PDF
Fuel Efficiency Trends	Line chart (cost/km over time)	Vehicle, fuel type	CSV

Report Type	Chart Type	Filters	Export
Maintenance Cost	Donut (by vehicle) + bar (by month)	Vehicle, service type	CSV, PDF
Compliance Status	Status matrix: license, insurance, permit per vehicle	Expiry window filter	PDF

8. Navigation & Information Architecture

8.1 Sidebar Navigation Structure

Nav Item	Icon	Roles with Access	Sub-items
Command Center	layout-dashboard	All roles	None
Vehicles	truck	Admin, Ops Manager	Registry, Documents, Maintenance
Trips	map-pin	Admin, Ops Manager, Dispatcher	Active Trips, History, Dispatch
Drivers	users	Admin, Ops Manager, Safety Officer	Profiles, Performance, Incidents
Fuel & Expenses	dollar-sign	Admin, Accountant	Fuel Logs, Expense Tracker
Rules Engine	zap	Admin only	Active Rules, Rule Builder, Alert Log
Analytics	bar-chart-2	Admin, Ops Manager, Accountant	Reports, Export Center
Settings	settings	Admin only	Company, Users, Roles, Integrations

8.2 Role-Based Visibility

The sidebar dynamically renders only the modules each user role has access to. Attempting to access a restricted URL redirects to a "403 Access Restricted" page with the user's current role displayed and a link to request access from Admin.

9. Interaction & Motion Design

Interaction	Animation	Duration	Easing
Page transition	Fade + slight upward translate (8px)	200ms	ease-out
Modal open	Scale from 96% + fade in	200ms	ease-out
Sidebar collapse/expand	Width transition	240ms	ease-in-out
Right panel slide-in	TranslateX from +100%	280ms	cubic-bezier(0.16,1,0.3,1)
Status pill change	Color crossfade	400ms	ease-in-out
Toast notification	Slide in from top-right + auto-exit	300ms	ease-out
Table row hover	Background color transition	120ms	linear
Button press	Scale to 0.97 on active	80ms	ease
KPI counter update	Animated count-up from previous value	600ms	ease-out
Alert pulse	Ring pulse animation on critical alerts	Loop 2s	ease-in-out

All animations respect prefers-reduced-motion: reduce. When this OS setting is active, transitions are instant (0ms) except for opacity fades which remain at 150ms for essential visual feedback.

10. Responsive Breakpoints

Breakpoint	Min Width	Layout Changes	KPI Grid
Mobile S	320px	Single column, sidebar hidden (hamburger)	1 column
Mobile L	480px	Single column, bottom nav tab bar	1 column
Tablet	768px	Sidebar auto-collapsed to icon rail, 8-col grid	2 columns
Desktop S	1024px	Full sidebar (260px), 12-col grid	3 columns
Desktop L	1280px	Full sidebar + optional right panel	4 columns
Ultra-wide	1920px+	Content max-width: 1440px, centered	4 columns (wider cards)

The Driver mobile view is a simplified read-only interface for trip logging and incident submission. Full fleet management features are desktop-optimized. Phase 2 will introduce a dedicated native mobile app for drivers.

11. Accessibility Standards

11.1 WCAG 2.1 AA Compliance

Criterion	Requirement	FEETFLOW Implementation
Color Contrast	Minimum 4.5:1 for normal text	Electric (#00C2FF) on navy (#0A0F1E) = 8.2:1 ✓
Focus Indicators	Visible focus ring on all interactive elements	2px solid electric, 2px offset
Status Information	Never convey state by color alone	All status pills include icon + text label
Keyboard Navigation	Full app operable by keyboard	Tab order follows visual flow, skip-to-main link
Error Identification	Errors described in text	Inline error messages with specific resolution guidance
Touch Targets	Minimum 44×44px	All buttons and icons meet minimum (48px preferred)

12. Design Handoff Checklist

Figma Deliverables

Deliverable	Status	Owner
Component Library (all states)	Required	UI Designer
Design Token file (JSON)	Required	UI Designer + Frontend Dev
All 8 screen flows (desktop)	Required	UI Designer
Mobile breakpoint variants	Required	UI Designer
Prototype with interactions	Recommended	UI Designer
Accessibility annotation layer	Required	UI Designer
Icon set (SVG export)	Required	UI Designer
Motion spec document	Required	UI Designer

Development Handoff

Item	Details
Tech Stack	React / Next.js + Tailwind CSS + Chart.js / Recharts
Component Framework	Tailwind UI base components, customized to FEETFLOW tokens
Icon Library	Lucide Icons (consistent with Figma assets)
Chart Library	Recharts for dashboards, Chart.js for analytics reports
Animation Library	Framer Motion for page transitions and micro-interactions
Design Tokens	CSS custom properties + Tailwind config extension
Dark Mode	Default (always dark), light mode for report exports only
Font Loading	Inter via Google Fonts with font-display: swap